

1

i

Our goal is to show that the edges in T^* are all the same (although the edge weights may be different).

If T is an optimal message-passing tree then T is an MST of a graph whose edge weights are $w_e = p_e$

If T is a MST of a graph whose edge weights are $w_e = p_e$ then T is an optimal message-passing tree

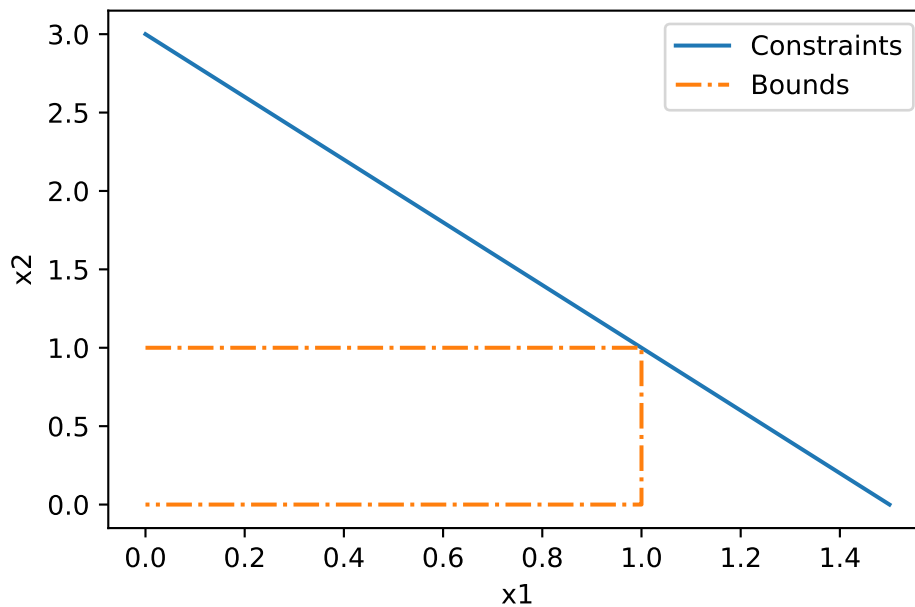
If T is not an optimal message-passing tree then T is not a MST of a graph whose edge weights are $w_e = p_e$

If T is not a MST of a graph whose edge weights are $w_e = p_e$ then T is not an optimal message-passing tree

ii

2

a



This program has a unique optimal solution which is $x^* = (x_1 = 1, x_2 = 1)$

b

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$c = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$b = (3)$$

c

```
#| echo: true

Maximize
  obj: x1 + x2
Subject To
  c1: 2x1 + x2 <= 3
Bounds
```

3

a

b

c

d

4

5

a

b

c

d

e

6

a

b

c

d

e

f